

PAIR OF STRAIGHT LINES

Synopsis-1

PAIR OF STRAIGHT LINES

1. Let us consider the equation:

$$3x^2 + 7xy - 6y^2 + 5x - y - 2 = 0 \quad \dots(1)$$

This equation can be written as $(3x - 2y - 1)(x + 3y + 2) = 0$

Clearly the equation (1) is satisfied by the coordinates of all the points which satisfy $3x - 2y - 1 = 0$ and also by the coordinates of all the points which satisfy $x + 3y + 2 = 0$ (2) But the equation (2) and (3) represent straight lines. Hence the locus represented by (1) is a pair of straight lines.

From the above example it can be observed that any equation of second degree in x any y represents a pair of straight lines if the expression in it is resolvable into two linear factors.

2. The homogeneous equation of the second degree $ax^2 + 2hxy + by^2 = 0$, $h^2 \geq ab$ represents a pair of straight lines passing through the origin.

Proof. Given equation is $ax^2 + 2hxy + by^2 = 0$

Case-I. Let $a \neq 0$

Multiplying the equation (1) by a, we have $a^2x^2 + 2ahxy + aby^2 = 0$

$$\text{i.e. } a^2x^2 + 2ahxy + h^2y^2 - h^2y^2 + aby^2 = 0 \Rightarrow (ax + hy)^2 - (h^2 - ab)y^2 = 0, (h^2 - ab > 0)$$

$$\Rightarrow (ax + hy)^2 - \left[y\sqrt{(h^2 - ab)} \right]^2 = 0 \Rightarrow \left[ax + hy + y\sqrt{(h^2 - ab)} \right] \left[ax + hy - y\sqrt{(h^2 - ab)} \right] = 0.$$

$\therefore$ The equation (1) represents the two straight lines

$$ax + hy + y\sqrt{(h^2 - ab)} = 0 \quad \dots(2) \qquad ax + hy - y\sqrt{(h^2 - ab)} = 0 \quad \dots(3)$$

Each of these lines are satisfied by (0, 0). Hence (2) and (3) pass through the origin.

Case-II. Let $a = 0$

$$\text{The given equation (1) reduces to } 2hxy + by^2 = 0 \Rightarrow 2(2hx + by) = 0$$

Now the two lines are $y = 0$ and $2hx + by = 0$.

These two equations represent two lines passing through the origin.

Note 1. If $h^2 > ab$, then the two lines are distinct.

2. If $h^2 = ab$, the two lines are coincident.

3. Corollary: if $ax^2 + 2hxy + by^2 = 0$ represents the pair of lines $l_1x + m_1y = 0$ and $l_2x + m_2y = 0$, then $l_1l_2 = a$, $m_1m_2 = b$ and $l_1m_2 + l_2m_1 = 2h$.

Proof. The two lines $l_1x + m_1y = 0$ and $l_2x + m_2y = 0$ are represented by $ax^2 + 2hxy + by^2 = 0$

$$\begin{aligned} \Leftrightarrow ax^2 + 2hxy + by^2 &= (l_1x + m_1y)(l_2x + m_2y) \\ &= l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 \end{aligned}$$

Equating the coefficients of x^2 , y^2 and xy on both sides

$$l_1 l_2 = a, m_1 m_2 = b \text{ and } l_1 m_2 + l_2 m_1 = 2h.$$

Corollary. If $ax^2 + 2hxy + by^2 = 0$ represents the pair of lines $y = m_1x$ and $y = m_2x$

$$\text{then } m_1 + m_2 = -\frac{2h}{b} \text{ and } m_1 m_2 = \frac{a}{b} (b \neq 0).$$

Proof. The two lines $y - m_1x = 0$ and $y - m_2x = 0$ are represented by $ax^2 + 2hxy + by^2 = 0$

$$\Rightarrow b(y - m_1x)(y - m_2x) \equiv ax^2 + 2hxy + by^2.$$

Equating the coefficients of xy and x^2 on both side

$$-b(m_1 + m_2) = 2h \text{ and } bm_1 m_2 = a \Rightarrow m_1 + m_2 = -\frac{2h}{b} m_1 m_2 \frac{a}{b}$$

4. Notation. The following notation will be adopted in this chapter.

$$\text{I. } s \equiv ax^2 + 2hxy + by^2, \quad s_1 \equiv axx_1 + h(xy_1 + x_1y) + byy_1$$

$$s_{11} \equiv ax_1^2 + 2hx_1y_1 + by_1^2, \quad s_{12} \equiv ax_1x_2 + h(x_1y_2 + x_2y_1) + by_1y_2$$

$$\text{II. } s \equiv ax^2 + 2hxy + by^2 + 2gx + 2fx + c$$

$$s_1 \equiv axx_1 + h(xy_1 + x_1y) + byy_1 + g(x + x_1) + f(y + y_1) + c$$

$$s_{11} \equiv ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c$$

$$s_{12} \equiv ax_1x_2 + h(x_1y_2 + x_2y_1) + by_1y_2 + g(x_1 + x_2) + f(y_1 + y_2) + c$$

5. An angle between the pair of lines $ax^2 + 2hxy + by^2 = 0$ is

$$\theta = \cos^{-1} \left(\frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} \right).$$

Proof. Let the lines be

$$l_1x + m_1y = 0 \quad \dots(1) \quad l_2x + m_2y = 0 \quad \dots(2)$$

$$\therefore ax^2 + 2hxy + by^2 \equiv (l_1x + m_1y)(l_2x + m_2y) = l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2$$

Equation the coefficients of like terms $l_1l_2 = a, l_1m_2 + l_2m_1 = 2h, m_1m_2 = b.$

$$\text{If } \theta = \frac{l_1l_2 + m_1m_2}{\sqrt{(l_1^2 + m_1^2) \cdot (l_2^2 + m_2^2)}} = \frac{l_1l_2 + m_1m_2}{\sqrt{l_1^2l_2^2 + l_1^2m_2^2 + l_2^2m_1^2 + m_1^2m_2^2}}$$

$$= \frac{l_1l_2 + m_1m_2}{\sqrt{(l_1^2m_2^2 + l_2^2m_1^2 - 2l_1l_2m_1m_2) + (l_1^2l_2^2 + m_1^2m_2^2 + 2l_1l_2m_1m_2)}}$$

$$= \frac{l_1l_2 + m_1m_2}{\sqrt{(l_1m_2 + l_2m_1)^2 + [(l_1l_2 + m_1m_2)^2]}} = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}}$$

$$\therefore \theta = \cos^{-1} \left(\frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} \right)$$

Note:

- The acute angle θ between the lines is given by $\cos \theta = \frac{|a+b|}{\sqrt{(a-b)^2 + 4h^2}}$.
- The angles between the pair of lines are $\theta, \pi - \theta$.
- The lines are coincident $\Leftrightarrow \theta = 0 \Leftrightarrow \cos \theta \Leftrightarrow \frac{|a+b|}{\sqrt{(a-b)^2 + 4h^2}} = 1$.
 $\Leftrightarrow (a-b)^2 + 4h^2 = (a+b)^2 \Leftrightarrow 4h^2 = (a+b)^2 - (a-b)^2 \Leftrightarrow h^2 = ab$.
- The lines are perpendicular $\Leftrightarrow \theta = 90^\circ \Leftrightarrow \cos \theta = \cos 90^\circ$
 $\Leftrightarrow \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}} = 0 \Leftrightarrow a+b=0 \Leftrightarrow \text{coeff. of } x^2 + \text{coeff. of } y^2 = 0$.
- $\tan \theta = \frac{2\sqrt{h^2 - ab}}{a+b}, \sin \theta = \frac{\sqrt{h^2 - ab}}{\sqrt{(a-b)^2 + 4h^2}}$.

Worksheet -1**PAIR OF STRAIGHT LINES****LEVEL-1****SINGLE CORRECT CHOICE TYPE:**

- The equation to the pair of lines represented by the equations $2x + y = 0$, $x + 2y = 0$ is
 - $2x^2 + 5xy + 2y^2 = 0$
 - $2x^2 + 5xy - 2y^2 = 0$
 - $2x^2 - 5xy + 2y^2 = 0$
 - $2x^2 - 5xy - 2y^2 = 0$
- The lines represented by the equation $3x^2 - 5xy + 2y^2 = 0$ are
 - $x - y = 0, 3x + y = 0$
 - $x + y = 0, x - y = 0$
 - $x - y = 0, 3x - 2y = 0$
 - $x + y = 0, 3x - y = 0$
- The angle between the straight lines $x^2 + 4xy + y^2 = 0$ is
 - 30°
 - 45°
 - 60°
 - 90°
- The sum of the slopes of the lines represented by $6x^2 - 5xy + y^2 = 0$ is
 - 5
 - 6
 - 7
 - 8

MULTI CORRECT CHOICE TYPE:

- The lines represented by the equation $6x^2 - xy + 4cy^2 = 0$ is $3x + 4y = 0, kx - 3y = 0$, then
 - $c = -3$
 - $c = 3$
 - $k = 2$
 - $k = -2$

REASONING TYPE:-

6. Statement I : The angle between the pair of lines $6x^2 - 5xy - 6y^2 = 0$ is $\frac{\pi}{2}$

Statement II : If θ is an angle between the lines represented by $ax^2 + 2hxy + by^2 = 0$,

$$\text{then } \cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}}$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

If ' θ ' is the angle between the lines represented by $ax^2 + 2hxy + by^2 = 0$, then

$$\cos \theta = \frac{a+b}{\sqrt{(a-b)^2 + 4h^2}}$$

7. If θ is the angle between the lines represented by $x^2 + 3xy - 4y^2 = 0$, then $\cos \theta =$

- 1) $\frac{\pi}{6}$ 2) $\frac{\pi}{3}$ 3) $\frac{-5}{\sqrt{34}}$ 4) $\frac{-3}{\sqrt{34}}$

8. The angle between the lines represented by $2x^2 - 7xy + 3y^2 = 0$ is

- 1) $\frac{\pi}{2}$ 2) $\frac{\pi}{3}$ 3) $\frac{\pi}{4}$ 4) $\frac{\pi}{6}$

9. The angle between the lines represented by $6x^2 + 11xy + 3y^2 = 0$, then $\tan \theta =$

- 1) $\frac{9}{7}$ 2) $\frac{7}{9}$ 3) $\frac{3}{7}$ 4) $\frac{7}{3}$

MATRIX MATCH TYPE:

10. If $y = m_1x$ and $y = m_2x$ are the lines represented by the equation $ax^2 + 2hxy + by^2 = 0$, then match the following

COLUMN -I

COLUMN -II

1) $m_1 + m_2 =$

p) $\frac{2\sqrt{h^2 - ab}}{b}$

2) $\frac{1}{m_1} + \frac{1}{m_2} =$

q) $\frac{-2h}{b}$

$$3) m_1 \times m_2 = \quad \quad \quad r) \frac{-2h}{a}$$

$$4) m_1 - m_2 = \quad \quad \quad s) \frac{a}{b}$$

LEVEL-2**SINGLE CORRECT CHOICE TYPE:**

11. If the sum of the slopes of the lines given by $x^2 - 2cxy - 7y^2 = 0$ is four times their product, then 'c' has the value

- 1) 1 2) -2 3) 2 4) -1

Multi Correct Choice Type:

12. If m_1, m_2 are the slopes of the lines $3x^2 - 8xy - 3y^2 = 0$ then

- 1) $m_1 + m_2 = \frac{-8}{3}$ 2) $m_1 \times m_2 = -1$ 3) $m_1 - m_2 = \frac{-10}{3}$ 4) $\frac{m_1}{m_2} = 0$

LEVEL-3**SINGLE CORRECT CHOICE TYPE:**

13. The equation to the two lines represented by the equation $x^2 + 2xy \cos \theta + y^2 = 0$ is

- 1) $x + y(\cos \theta + \cot \theta) = 0, x + y(\cos \theta - \cot \theta) = 0$
 2) $x + y(\cos \theta) = 0, x + y(\cot \theta) = 0$
 3) $x + y = 0, x - \sin \theta = 0$
 4) $x - \sec \theta = 0, x + \cot \theta = 0$

MATRIX MATCH TYPE:

14. If ' θ ' is the angle between the pair of lines then match the following

COLUMN -I

COLUMN -II

- | | |
|--|-------------------------|
| 1) If $2x^2 + 5xy + 2y^2 = 0$, then $\tan \theta =$ | p) $\frac{3}{4}$ |
| 2) If $2x^2 - 5xy + 2y^2 = 0$, then $\cos \theta =$ | q) $\frac{4}{5}$ |
| 3) If $xy + y^2 = 0$, then $\sin \theta =$ | r) $\frac{1}{\sqrt{2}}$ |
| 4) If $x^2 - 7xy + 12y^2 = 0$, then $\frac{2 \cos \theta + 3 \sin \theta}{4 \sin \theta + 5 \cos \theta}$ | s) $\frac{39}{69}$ |

LEVEL-4

SINGLE CORRECT CHOICE TYPE:

15. The combined equation to a pair of straight lines passes through the origin and inclined 30° and 60° respectively with x-axis is

1) $\sqrt{3}(x^2 + y^2) = 4xy$

2) $4(x^2 + y^2) = \sqrt{3}xy$

3) $x^2 + \sqrt{3}y^2 - 2xy = 0$

4) $x^2 + 3y^2 - 2xy = 0$

REASONING TYPE:

16. *Statement I* : If the equation $4x^2 + mxy - 3y^2 = 0$ represents a pair of real and distinct lines then $m \in \mathbb{R}$.

Statement II : If the difference of the slopes of the lines $kx^2 - 12xy + y^2 = 0$ is 2 then k is 3.

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

LEVEL-5**SINGLE CORRECT CHOICE TYPE:**

17. If $\theta_1, \theta_2, \theta_3$ are the angles between the pairs of lines $2x^2 + 3xy - 2y^2 = 0$, $x^2 + 4xy + y^2 = 0$, $xy - y^2 = 0$ then the ascending order of $\theta_1, \theta_2, \theta_3$ is

1) $\theta_2, \theta_1, \theta_3$

2) $\theta_3, \theta_2, \theta_1$

3) $\theta_1, \theta_2, \theta_3$

4) $\theta_1, \theta_3, \theta_2$

COMPREHENSION TYPE:

A : $2x^2 - 20xy + 8y^2 = 0$;

B : $-2x^2 + 4xy - 3y^2 = 0$

C : $ab(x^2 - y^2) + (a^2 - b^2)xy = 0$

D : $x^2 + 2xy \sec \theta + y^2 = 0$

A, B, C and D represent a pair of two straight lines each passing through the origin.

18. The equation of pairs of straight line represented $A - B = 0$ are

1) $(2x - y)(2x - 11y) = 0$

2) $(2x + y)(2x + 11y) = 0$

3) $(2x - y)(2x + 11y) = 0$

4) $(2x + y)(2x - 11y) = 0$

19. The equations of pair of straight lines represented by C are

1) $ax - by = 0$ or $bx - ay = 0$

2) $ax + by = 0$ or $bx - ay = 0$

3) $ax - by = 0$ or $bx + ay = 0$

4) $ax + by = 0$ or $bx + ay = 0$

20. The equations of the lines whose joint equation is $x^2 + 2xy \sec \theta + y^2 = 0$ are

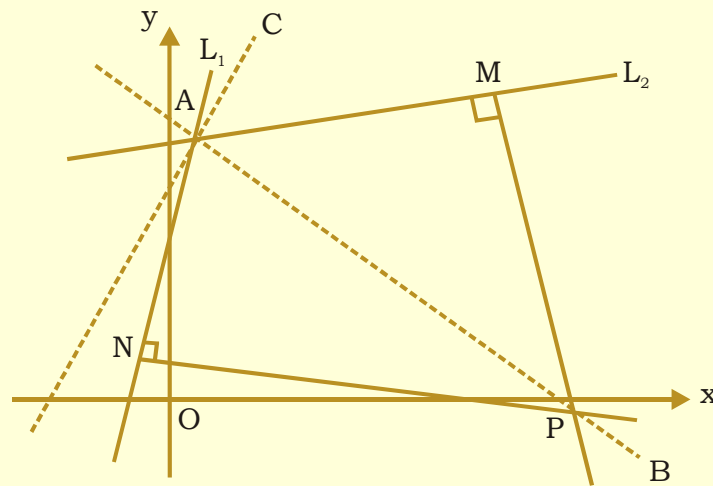
- 1) $x(\tan \theta - \sec \theta)$ and $x(\tan \theta - \sec \theta)$
- 2) $x(\tan \theta - \sec \theta)$ and $x(\sec \theta + \tan \theta)$
- 3) $x(\tan \theta - \sec \theta)$ and $x(-\sec \theta - \tan \theta)$
- 4) $x(\tan \theta + \sec \theta)$ and $x(-\sec \theta - \tan \theta)$

Synopsis -2

PAIR OF STRAIGHT LINES

1. Lines bisecting the angles between two lines

Corollary: $L_1 \equiv a_1x + b_1y + c_1 = 0$, $L_2 \equiv a_2x + b_2y + c_2 = 0$ and $c_1c_2 > 0$. Equation to the line bisecting the angle containing the origin between the lines L_1 , L_2 is



$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = + \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

and to the line bisecting the other angle between the lines L_1 , L_2 is

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = - \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

Proof. Equations L_1 , L_2 are $a_1x + b_1y + c_1 = 0$ (1)
 $a_2x + b_2y + c_2 = 0$ (2)

and $c_1c_2 > 0$.

We know that if P is any point on one of the lines bisecting the angles between L_1 , L_2 , then the perpendicular distances of P from L_1 , L_2 are equal. $P(x, y)$ is any point on the bisecting lines $\Leftrightarrow$ distance of P from L_1 = distance of P from L_2 .

$$\Leftrightarrow \left| \frac{a_1x + b_1y + c_1}{\sqrt{(a_1^2 + b_1^2)}} \right| = \left| \frac{a_2x + b_2y + c_2}{\sqrt{(a_2^2 + b_2^2)}} \right| \Leftrightarrow \frac{a_1x + b_1y + c_1}{\sqrt{(a_1^2 + b_1^2)}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{(a_2^2 + b_2^2)}}$$

$$\therefore \frac{a_1x + b_1y + c_1}{\sqrt{(a_1^2 + b_1^2)}} = + \frac{a_2x + b_2y + c_2}{\sqrt{(a_2^2 + b_2^2)}} \text{ and } \frac{a_1x + b_1y + c_1}{\sqrt{(a_1^2 + b_1^2)}} = - \frac{a_2x + b_2y + c_2}{\sqrt{(a_2^2 + b_2^2)}}$$

are the lines bisecting the angles between L_1, L_2 .

2. Corollary: the equation of the pair of bisectors of the angles between the pair of lines $ax^2 + 2hxy + by^2 = 0$ is $h(x^2 - y^2) - xy(a - b) = 0$.

Proof: Let the euquation $ax^2 + 2hxy + by^2 = 0$ (1)

represent the two lines taken in symmetric form $x \sin \alpha \ y \cos \alpha = 0$ (2)

and $x \sin \beta - y \cos \beta = 0$ (3)

Then for some non-zero $\lambda \in \mathbb{R}$

$$\begin{aligned} ax^2 + 2hxy + by^2 &= \lambda(x \sin \alpha - y \cos \alpha)(x \sin \beta - y \cos \beta) \\ &= \lambda x^2 \sin \alpha \sin \beta - \lambda xy \sin(\alpha + \beta) + \lambda y^2 \cos \alpha \cos \beta. \end{aligned}$$

$$\Rightarrow \frac{a}{\lambda} = \sin \alpha \sin \beta; \quad \frac{-2h}{\lambda} = \sin(\alpha + \beta); \quad \frac{b}{\lambda} = \cos \alpha \cos \beta.$$

Now the equations to the lines bisecting the angles between the lines (2) and (3) are

$$\frac{x \sin \alpha - y \cos \alpha}{\sqrt{\sin^2 \alpha + \cos^2 \alpha}} = \pm \frac{x \sin \beta - y \cos \beta}{\sqrt{\sin^2 \beta + \cos^2 \beta}} \quad \text{.....(I)}$$

$$\therefore \text{ The bisectors are } (x \sin \alpha - y \cos \alpha) + (x \sin \beta - y \cos \beta) = 0 \quad \text{.....(A)}$$

$$(x \sin \alpha - y \cos \alpha) - (x \sin \beta - y \cos \beta) = 0 \quad \text{.....(B)}$$

$\therefore$ Equation to the pair of angle bisectors is the product of (A) and (B)

$$\Rightarrow (x \sin \alpha - y \cos \alpha)^2 - (x \sin \beta - y \cos \beta)^2$$

$$\Rightarrow x^2(\sin^2 \alpha - \sin^2 \beta) - xy(\sin 2\alpha - \sin 2\beta) + y^2(\cos^2 \alpha - \cos^2 \beta) = 0$$

$$\Rightarrow (x^2 - y^2)(\sin^2 \alpha - \sin^2 \beta) = xy(\sin 2\alpha - \sin 2\beta)$$

$$\Rightarrow (x^2 - y^2)\sin(\alpha + \beta)\sin(\alpha - \beta) = xy[2\cos(\alpha + \beta)\sin(\alpha - \beta)]$$

$$\Rightarrow (x^2 - y^2)\sin(\alpha + \beta) = 2xy\cos(\alpha + \beta)$$

$$\Rightarrow (x^2 - y^2)\left(\frac{-2h}{\lambda}\right) = (2xy)\left[\frac{b}{\lambda} - \frac{a}{\lambda}\right] \Rightarrow h(x^2 - y^2) = (a - b)xy.$$

Note: If $a \neq b$, $h \neq 0$, then the equation to the pair bisectors can be written as

$$\frac{x^2 - y^2}{a - b} = \frac{xy}{h}.$$

3. Equally Inclined Pair of Lines

Definition: If the lines $L_1 = 0$ and $L_2 = 0$ makes equal angles with the lines $L = 0$, then the pair of lines $L_1 L_2 = 0$ is said to be equally inclined with the line $L = 0$.

Note: Every pair of lines is equally inclined to each of its angle bisectors.

Definition. Two pairs of lines are said to be equally inclined to each other if they possess the same pair of angle bisectors.

Ex.1. Find the acute angle between the pair of lines * (a) $3x^2 - 5xy - 2y^2 = 0$.

* (b) $2x^2 - 3xy - 6y^2 = 0$ * (c) $(x \cos \alpha - y \sin \alpha)^2 = (x^2 + y^2) \sin^2 \alpha$

Sol: Comparing with the equation $ax^2 + 2hxy + by^2 = 0$.

We have $a = 3$, $b = -2$, $2h = -5$

$\therefore$ The acute angle θ between the given pair of lines is given by

$$\cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + 4h^2}} = \frac{3 + 2}{\sqrt{(3 - 2)^2 + (-5)^2}} = \frac{1}{5\sqrt{2}} \Rightarrow \theta = \cos^{-1} \left(\frac{1}{5\sqrt{2}} \right)$$

(b) From the equation $2x^2 - 3xy - 6y^2 = 0$. We have $a = 2$, $2h = -3$, $b = -6$

$\therefore$ The acute angle θ between the given pair of lines is given by

$$\cos \theta = \left| \frac{2 - 6}{\sqrt{(2 - 6)^2 + (-3)^2}} \right| = \frac{4}{\sqrt{73}} \Rightarrow \theta = \cos^{-1} \left(\frac{4}{\sqrt{73}} \right).$$

(c) From the equation $(x \cos \alpha - y \sin \alpha)^2 = (x^2 + y^2) \sin^2 \alpha$

$$\Rightarrow x^2 (\cos^2 \alpha - \sin^2 \alpha) - 2xy \sin \alpha \cos \alpha = 0 \Rightarrow x^2 \cos 2\alpha - xy \sin 2\alpha = 0$$

$\therefore$ The acute angle θ between the lines is given by

$$\cos \theta = \frac{\cos 2\alpha}{\sqrt{(\cos 2\alpha)^2 + \sin^2 2\alpha}} \Rightarrow \cos \theta = \cos 2\alpha \Rightarrow \theta = 2\alpha.$$

Corollary: The area of the triangle formed by $ax^2 + 2hxy + by^2 = 0$ and

$$lx + my + n = 0 \text{ is } \left| \frac{n^2 \sqrt{(h^2 - ab)}}{am^2 - 2hlm + bl^2} \right|.$$

Proof. Let the line given by $ax^2 + 2hxy + by^2 = 0$ be $p_1x + q_1y = 0$ (1)

$p_2x + q_2y = 0$ (2). Then $p_1p_2 = a, q_1q_2 = b, p_1q_2 + p_2q_1 = 2h$. Given third line is $lx + my + n = 0$ (3)

Let (1) and (2) intersect (3) in $A(x_1, y_1)$ and $B(x_2, y_2)$. Then $p_1x_1 + q_1y_1 = 0, lx_1 + my_1 + n = 0$

$$\text{Solving : } \frac{x_1}{nq_1} = \frac{y_1}{0 - np_1} = \frac{1}{mp_1 - lq_1} \Rightarrow A(x_1, y_1) = \left(\frac{nq_1}{mp_1 - lq_1}, \frac{np_1}{mp_1 - lq_1} \right)$$

$$\text{Similarity } B(x_2, y_2) = \left(\frac{nq_2}{mp_2 - lq_2}, \frac{-np_2}{mp_2 - lq_2} \right) \therefore \text{Area of } \triangle OAB = \frac{1}{2} |x_1y_2 - x_2y_1|$$

$$\Rightarrow \text{Area } \triangle OAB = \frac{1}{2} \left| \frac{-n^2 p_2 q_1 + n^2 p_1 q_2}{(mp_1 - lq_1)} \right| = \left| \frac{n^2 (p_1 q_2 - p_2 q_1)}{2 [p_1 p_2 m^2 - (p_1 q_2 + p_2 q_1) ml + q_1 q_2 l^2]} \right|$$

$$\Rightarrow \left| \frac{n^2 \sqrt{(p_2 q_1 + p_1 q_2)^2 - 4 p_1 p_2 q_1 q_2}}{2 [am^2 - 2hlm + bl^2]} \right| = \left| \frac{n^2 \sqrt{4h^2 - 4ab}}{2(am^2 - 2hlm + bl^2)} \right| = \left| \frac{n^2 \sqrt{h^2 - ab}}{am^2 - 2hlm + bl^2} \right|$$

Ex. Find the area of the triangle formed by the pair of lines $x^2 + 4xy + y^2 = 0$ and $x + y - 1 = 0$.

Sol. From given equation of pair of lines, $a = 1, 2h = 4, b = 1$

$$\therefore \text{Area of the triangle} = \frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|} = \frac{(-1) \sqrt{(2)^2 - (1)(1)}}{1(1)^2 - 4(1)(1) - 4(1)(1) + 1(1)^2}$$

$$= \frac{\sqrt{3}}{2} \text{ sq. units}$$

Worksheet -2

PAIR OF STRAIGHT LINES

LEVEL-1

SINGLE CORRECT CHOICE TYPE:

- The angle bisectors of the coordinate axes are
 - $x + 1 = 0, y = 0$
 - $x + y = 0, x - y = 0$
 - $x + 1 = 0, y + 1 = 0$
 - $x + y - 1 = 0, x - y - 5 = 0$
- The angle bisectors of the lines $x + y + 1 = 0, x - y + 1 = 0$ are
 - $x + 1 = 0, y = 0$
 - $x + y = 0, x - y = 0$
 - $x + 1 = 0, y + 1 = 0$
 - $x + y + 1 = 0, x - y - 5 = 0$
- The equation of the bisectors of the acute angle between the lines $3x - 4y + 7 = 0$

and $5y + 12x - 2 = 0$ is

- 1) $5x + 5y + 3 = 0$ 2) $10x - 3y + 7 = 0$ 3) $5x - 2y + 1 = 0$ 4) $11x - 3y + 9 = 0$

4. The area of the triangle formed by the lines $x^2 + 4xy + y^2 = 0$, $x + y = 1$ is

- 1) $\sqrt{3}$ sq.units 2) 2sq.units 3) 1sq.units 4) $\frac{\sqrt{3}}{2}$ sq.units

MULTI CORRECT CHOICE TYPE:

5. The equation of the pair of bisectors of the angle between the pair of lines $x^2 - 2axy - y^2 = 0$ is $x^2 - 2bxy - y^2 = 0$, then

- 1) $ab = -1$ 2) $ab = 1$ 3) $ab + 1 = 0$ 4) $ab = 2$

REASONING TYPE:

6. *Statement I* : The area of the triangle formed by the lines $12x^2 + 20xy + 7y^2 = 0$, $2x - 3y + 4 = 0$ is 4sq.units.

Statement II : The area of the triangle formed by $ax^2 + 2hxy + by^2 = 0$ and

$$lx + my + n = 0 \text{ is } \frac{n^2 \sqrt{h^2 - ab}}{|am^2 - 2hlm + bl^2|}.$$

1. Both Statements are true, Statement II is the correct explanation of Statement I.
2. Both Statements are true, Statement II is not correct explanation of Statement I.
3. Statement I is true, Statement II is false.
4. Statement I is false, Statement II is true.

COMPREHENSION TYPE:

The equations of the lines bisector of the angles between the lines

$$a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0 \text{ are } \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}} = 0.$$

7. The equations of the bisector of the angles between the lines $3x - 4y + 2 = 0$, $12x - 5y + 2 = 0$ is

- 1) $21x + 27y - 16 = 0$ 2) $99x + 77y + 36 = 0$
 3) $21x - 27y + 16 = 0$ 4) $99x + 77y - 36 = 0$

8. The equation to the pair of bisector of the angle between the pair of lines $2x^2 - 3xy + y^2 = 0$ is

- 1) $3x^2 + 2xy - 3y^2 = 0$ 2) $3x^2 - 2xy + 3y^2 = 0$
 3) $3x^2 + 2xy + 3y^2 = 0$ 4) $3x^2 - 2xy - 3y^2 = 0$

9. The equation of the bisectors of the angle between the two straight lines

$$2x^2 - 3xy + y^2 = 0 \text{ is}$$

1) $3x^2 + 2xy - 3y^2 = 0$

2) $2x^2 + 3xy - 2y^2 = 0$

3) $3x^2 - 2xy - 3y^2 = 0$

4) $2x^2 - 3xy - 2y^2 = 0$

MATRIX MATCH TYPE:

10. COLUMN -I

Sides of the triangle

a) $x^2 + 4xy + y^2 = 0, x + y = 1$

COLUMN -II

Area of the triangle

p) 4

b) $x^2 - 3xy + y^2 = 0, x + y + 1 = 0$

q) $\frac{1}{\sqrt{3}}$

c) $12x^2 - 20xy + 7y^2 = 0, 2x - 3y + 4 = 0$

r) $\frac{\sqrt{3}}{2}$

d) $(3x + 4y)^2 - 3(3y - 4x)^2 = 0, 11x + 4y + 5 = 0$

s) $\frac{1}{2\sqrt{5}}$

LEVEL-2

SINGLE CORRECT CHOICE TYPE:

11. The area of the triangle formed by the lines $y = x + c$, $ax^2 + 2hxy + by^2 = 0$ is

1) $\frac{c^2 \sqrt{h^2 - ab}}{|a + b + 2h|}$

2) $\frac{c^2 \sqrt{h^2 + ab}}{|a + b + 2h|}$

3) $\frac{c^2 \sqrt{h^2 - ab}}{(a + b)^2 + 2h}$

4) $\frac{c^2 \sqrt{h^2 - ab}}{(a - b)^2 + 2h}$

MATRIX MATCH TYPE:

12. COLUMN -I

Pair of straight lines

a) $3x^2 - 5xy + 2y^2 = 0$

COLUMN -II

Pair of angle bisectors

p) $5x^2 - 24xy - 5y^2 = 0$

b) $x^2 + 4xy + y^2 = 0$

q) $5x^2 + 2xy - 5y^2 = 0$

c) $2x^2 + 6xy + y^2 = 0$

r) $x^2 - y^2 = 0$

d) $6x^2 + 5xy - 6y^2 = 0$

s) $3x^2 - xy - 3y^2 = 0$

LEVEL-3**SINGLE CORRECT CHOICE TYPE:**

13. If the equation of the pair of bisectors of the angle between the pair of lines $3x^2 + xy + by^2 = 0$ is $x^2 - 14xy - y^2 = 0$, then $b =$
- 1) 4 2) - 4 3) 8 4) - 8
14. If the area of the triangle formed by the lines $3x^2 - 2xy - 8y^2 = 0$ and the line $2x + y - k = 0$ is 5sq.units, then $k =$
- 1) 5 2) 6 3) 7 4) 8

REASONING TYPE:

15. *Statement I* : The equations of the sides AB, BC, CA of triangle ABC are $3x + 4y = 6$, $12x - 5y = 3$ and $4x - 3y + 12 = 0$. The equation of the internal bisector of $\angle A$ is $7x + 7y + 18 = 0$.

Statement II : If $L_1 = a_1x + b_1y + c_1 = 0$; $L_2 = a_2x + b_2y + c_2 = 0$ and $c_1 c_2 > 0$ then equation to the line bisecting the angle containing the origin between the lines

$$L_1 \text{ and } L_2 \text{ is } \frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm \frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}.$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

LEVEL-4**Multi Correct Choice Type:**

16. The area of the triangle formed by two lines whose combined equation is $y = |x|$ and the line $x + 2y = 2$ is

- 1) $\frac{8}{3}$ sq.units 2) $\frac{4}{3}$ sq.units 3) 4sq.units 4) $\frac{16}{3}$ sq.units

MATRIX MATCH TYPE:

17. COLUMN -I

Sides of triangle

COLUMN -II

Type of triangle formed

- a) $x^2 - 4xy + y^2 = 0$ and $x + y = 3$ p) Acute angled isosceles
 b) $xy = 0$ and $x + y = 3$ q) Obtuse angled isosceles
 c) $x^2 + 4xy + y^2 = 0$, $x + y = 3$ r) Equilateral
 d) $(x + y)^2 = (7 + 4\sqrt{3})(x - y)^2$ and $x + y = 3$ s) Right angled isosceles

LEVEL-5**Multi Correct Choice Type:**

18. Two pairs of straight lines have the equations $y^2 + xy - 12x^2 = 0$ and $ax^2 + 2hxy + by^2 = 0$. One line will be common among them. if
 1) $a = -3(2h + 3b)$ 2) $a = 8(h - 2b)$ 3) $a = 2(b + h)$ 4) $a = -3(b + h)$
19. If one of the lines of $my^2 + (1 - m^2)xy - mx^2 = 0$ is a bisector of the angle between the lines $xy = 0$, then m is
 1) 1 2) 2 3) $-\frac{1}{2}$ 4) -1

Worksheet -3**PAIR OF STRAIGHT LINES**

Corollary: If the equation $S \equiv ax^2 + hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines then (i) $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 0$ and (ii) $h^2 \geq ab$, $g^2 \geq ac$, $f^2 \geq bc$

Proof : Let the equation $S = 0$ represent the lines

$$l_1x + m_1y + n_1 = 0; \quad l_2x + m_2y + n_2 = 0.$$

Hence the equation to pair of lines will be $(l_1x + m_1y + n_1)(l_2x + m_2y + n_2) = 0$

Equating the coefficients of like terms

$$l_1l_2 = a; \quad l_1m_2 + l_2m_1 = 2h; \quad m_1m_2 = b; \quad n_2 + l_2n_1 = 2g; \quad m_1n_2 + m_2n_1 = 2f; \quad n_1n_2 = c$$

(i) Consider the product $(2h)(2g)(2f) = (l_1n_2 + l_2n_1)(m_1n_2 + m_2n_1)(m_1n_2 + m_2n_1)$

$$= l_1l_2(m_1^2n_2^2 + m_2^2n_1^2) + m_1m_2(n_1^2l_2^2 + n_2^2l_1^2) + n_1n_2(l_1^2m_2^2 + l_2^2m_1^2) + 2l_1l_2m_1m_2n_1n_2$$

$$= l_1l_2[(m_1n_2 + m_2n_1) - 2m_1m_2n_1n_2] + m_2m_2[(n_1l_2 + n_2l_1)^2 - 2n_1n_2l_1l_2]$$

$$n_2n_2[(l_1m_2 + l_2m_1)^2 - 2l_1l_2m_1m_2] + 2l_1l_2m_1m_2n_1n_2$$

$$\therefore 8fgh = 4(af^2 + bg^2 + ch^2 - abc).$$

$$(ii) \text{ Consider } h^2 - ab = \left(\frac{l_1 m_2 + l_2 m_1}{2} \right) - l_1 l_2 m_1 m_2 = \frac{(l_1 m_2 - l_2 m_1)^2}{4} \geq 0$$

$\therefore h^2 - ab \geq 0 \Rightarrow h^2 \geq ab$. Similarly it can be proved that $g^2 \geq ac$ and $f^2 \geq bc$.

Note : Even though a second degree equation in x and y satisfies the condition $\Delta = 0$, it need not represent a pair of lines. Consider the equation, $x^2 + 2xy + y^2 + 1 = 0$. This does not represent a pair of lines since it cannot be factorised for $x, y \in \mathbb{R}$. Here $a = 1, b = 1, c = 1, h = 1, g = 0, f = 0$.

Clearly $\Delta = abc + 2fgh - af^2 - bg^2 - ch^2 = 1 - 1 = 0, h^2 \geq ab$.

But $g^2 < ac$ and $f^2 < bc$

$\therefore$ Even if one of the condition fails, the given equation does not represent a pair of lines.

Corollary: If $S \equiv ax^2 + hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines then the equation $ax^2 + 2hxy + by^2 = 0$ represents a pair of lines parallel to them through the origin.

Proof : Let $S = 0$ represents the two lines $l_1x + m_1y + n_1 = 0$ (1)

$$l_2x + m_2y + n_2 = 0 \quad \text{.....(2)}$$

$$\therefore ax^2 + 2hxy + by^2 + 2gx + 2fy + c \equiv (l_1x + m_1y + n_1)(l_2x + m_2y + n_2)$$

Equating the like terms on both sides $l_1l_2 = a, l_1m_2 + l_2m_1 = 2h$ and $m_1m_2 = b$

The equations of the lines parallel to (1) and (2) and passing through the origin are $l_1x + m_1y = 0, l_2x + m_2y = 0$.

The combined equation of parallel lines will be $(l_1x + m_1y)(l_2x + m_2y) = 0$

$$\Rightarrow l_1l_2x^2 + (l_1m_2 + l_2m_1)xy + m_1m_2y^2 = 0 \Rightarrow ax^2 + 2hxy + by^2 = 0$$

Note : The acute angle between the pair of lines $S = 0$ is equal to the acute angle between the pair of lines $ax^2 + 2hxy + by^2 = 0$.

Corollary: If (α, β) is the point of intersection of the pair of lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$, then the equation to the pair of bisectors of angles between them is $\left[h(x - \alpha)^2 - (y - \beta)^2 \right] = (a - b)(x - \alpha)(y - \beta)$.

Proof. The equation to the pair of lines through the origin and parallel to the given pair is $ax^2 + 2hxy + by^2 = 0$.

Equation to the pair of angle bisectors of (1) is $h(x^2 - y^2) = (a - b)xy$

∴ The equation to the pair of lines through (α, β) and parallel to (2) is

$[h(x - \alpha)^2 - (y - \beta)^2] = (a - b)(x - \alpha)(y - \beta)$ which is the required pair of angle bisectors.

Corollary: The point of intersection of the pair of lines represented by $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ when $h^2 > ab$ is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right)$.

Proof : Let the point of intersection of the given pair of lines be (x_1, y_1) . Transfer the origin to (x_1, y_1) without changing the direction of the axes.

Let (X, Y) represent the new coordinates of (x, y) . Then $x = X + x_1$ and $y = Y + y_1$

Now the given equation referred to new axes will be

$$a(X + x_1)^2 + 2h(X + x_1)(Y + y_1) + b(Y + y_1)^2 + 2g(X + x_1) + 2f(Y + y_1) + c = 0$$

$$\Rightarrow aX^2 + 2hXY + bY^2 + 2X(ax_1 + hx_1 + g) + 2Y(hx_1 + by_1 + f)$$

$$+ (ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c) = 0$$

Since this equation represents a pair of lines passing through the origin it should be homogeneous second degree equation in X and Y . Hence the first degree terms and the constant term must be zero.

$$ax_1 + hy_1 + g = 0 \quad \dots\dots(1) \quad \quad \quad hx_1 + by_1 + f = 0 \quad \dots\dots(2)$$

$$ax_1 + 2hx_1y_1 + by_1 + 2gx_1 + 2fy_1 + c = 0 \quad \dots\dots(3)$$

But (3) can be rearranged as

$$x_1(ax_1 + hy_1 + g) + y_1(hx_1 + by_1 + f) + (gx_1 + fy_1 + c) = 0$$

$$\Rightarrow gx_1 + fy_1 + c = 0 \quad \dots\dots(4)$$

$$\text{Solving (1) and (2) for } x_1 \text{ and } y_1 \quad \frac{x_1}{hf - bg} = \frac{y}{gh - af} = \frac{1}{ab - h^2}$$

$$\therefore x_1 = \frac{hf - bg}{ab - h^2} \text{ and } y_1 = \frac{gh - af}{ab - h^2} \quad (\because ab - h^2 \neq 0)$$

Hence the point of intersection of the given pair of lines is given by

$$\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2}\right).$$

Even by solving (1) and (4) or by solving (2) and (4) we get the point of intersection.

Corollary: If $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of parallel lines then $h^2 = ab$ and $bg^2 = af^2$. Also the distance between the two parallel

lines is $2\sqrt{\left[\frac{g^2 - ac}{a(a+b)}\right]}$.

Proof : Let the parallel lines represented by $S = 0$ be $lx + my + n_1 = 0$ (A)

$$lx + my + n_2 = 0 \quad \text{.....(B)}$$

$$\therefore ax^2 + 2hxy + by^2 + 2gx + 2fy + c = (lx + my + n_1)(lx + my + n_2) = 0$$

$$\text{Equating the like terms } l^2 = a \quad \text{.....(1)} \qquad 2lm = 2h \quad \text{.....(2)}$$

$$m^2 = b \quad \text{.....(3)} \qquad l(n_1 + n_2) = 2g \quad \text{.....(4)}$$

$$m(n_1 + n_2) = 2f \quad \text{.....(5)} \qquad n_1 + n_2 = c \quad \text{.....(6)}$$

From (1) and (3). $l^2m^2 = ab$, and from (2), $h^2 = ab$

$$\text{Dividing (4) and (5), } \frac{l}{m} = \frac{g}{f} \Rightarrow \frac{l^2}{m^2} = \frac{g^2}{f^2} \quad \therefore \frac{a}{b} = \frac{g^2}{f^2} \Rightarrow bg^2 = af^2$$

Distance between the parallel lines (A) and (B) is

$$\begin{aligned} &= \left| \frac{n_1 - n_2}{\sqrt{l^2 + m^2}} \right| = \frac{\sqrt{[(n_1 + n_2)^2 - 4n_1n_2]}}{\sqrt{l^2 + m^2}} = \frac{\sqrt{(4g^2/l^2) - 4c}}{\sqrt{(a+b)}} \\ &= \frac{\sqrt{(4g^2/a) - 4c}}{\sqrt{a+b}} = 2\sqrt{\left[\frac{g^2 - ac}{a(a+c)}\right]} \end{aligned}$$

Note : The distance between the parallel lines is also equal to $2\sqrt{\left[\frac{f^2 - bc}{b(a+b)}\right]}$.

Corollary: The product of the perpendiculars from a point (α, β) to the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } \left| \frac{a\alpha^2 + 2h\alpha\beta + b\beta^2}{\sqrt{(a-b)^2 + 4h^2}} \right|.$$

Proof : Let the equation $ax^2 + 2hxy + by^2 = 0$ represents the two lines

$$l_1x + m_1y = 0 \quad \text{.....(1)} \quad \text{and} \quad l_2x + m_2y = 0$$

Then $l_1 l_2 = a$, $m_1 m_2 = b$, $l_1 m_2 + l_2 m_1 = 2h$

$\therefore$ The lengths of perpendiculars from (α, β) to (1) and (2) are

$$\left| \frac{l_1 \alpha + m_1 \beta}{\sqrt{l_1^2 + m_1^2}} \right| \text{ and } \left| \frac{l_2 \alpha + m_2 \beta}{\sqrt{l_2^2 + m_2^2}} \right|$$

$$\therefore \text{Product of perpendiculars} = \left| \frac{l_1 l_2 \alpha^2 + (l_1 m_2 + l_2 m_1) \alpha \beta + m_1 m_2 \beta^2}{\sqrt{l_1^2 l_2^2 + m_1^2 m_2^2 + l_2^2 m_1^2 + l_1^2 m_2^2}} \right|$$

$$\Rightarrow \frac{|\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{((l_1 l_2 - m_1 m_2)^2 + (l_1 m_2 + l_2 m_1)^2)}} = \frac{|\alpha^2 + 2h\alpha\beta + b\beta^2|}{\sqrt{(a-b)^2 + 4h^2}}$$

Note. The product of the perpendiculars from a point (x_1, y_1) to the pair of lines

$$ax^2 + 2hxy + by^2 = 0 \text{ is } \frac{|ax_1^2 + 2hx_1y_1 + by_1^2|}{\sqrt{(a-b)^2 + 4h^2}} \text{ i.e., } \frac{|S_{11}|}{\sqrt{(a-b)^2 + 4h^2}}$$

Ex. Find the product of perpendiculars from $(1, -2)$ to the pair of lines

$$2x^2 - xy + y^2 = 0$$

$$\text{Sol. Product of the perpendiculars} = \frac{2(1) - (1)(-2) + (-2)^2}{\sqrt{(2-1)^2 + (-1)^2}} = 4\sqrt{2}$$

Corollary: The area of the triangle formed and is .

Proof. Let the line given by be(1)

.....(2). Then . Given third line is(3)

Let (1) and (2) intersect (3) in and . Then Solving : Similary

Ex. Find the area of the triangle formed by the pair of lines and .

Sol. From given equation of pair of lines, $a = 1$, $2h = 4$, $b = 1$

Corollary: The equation to pair of lines through the origin and forming an equilateral triangle with the line $ax + by + c = 0$ is $(ax + by)^2 - 3(bx - ay)^2 = 0$.

Proof. Given lines is $ax + by + c = 0$.

The equation to the line $\overleftrightarrow{OD}$ through an equilateral triangle with (1)

$$\Rightarrow \angle(\overleftrightarrow{OA}, \overleftrightarrow{OD}) = 30^\circ \quad \text{Let the equation to } \overleftrightarrow{OA} \text{ be } y = mx \quad \text{.....(3)}$$

$\therefore$ The angle between $bx - ay = 0$ and $mx - y = 0$ is 30°

$$\Rightarrow \cos 30^\circ = \left| \frac{(bm + a)}{\sqrt{(a^2 + b^2)(m^2 + 1)}} \right| \quad \text{Eliminating } m \text{ using (3) we have}$$

$$\frac{\sqrt{3}}{2} = \frac{\left(\frac{by}{x} + a\right)}{\sqrt{(a^2 + b^2)\left(\frac{y^2}{x^2} + 1\right)}} \quad \text{Squaring and cross multiplying}$$

$$4(by + ax)^2 = 3(a^2 + b^2)(x^2 + y^2) \Rightarrow x^2(a^2 - 3b^2) + 8abxy + y^2(b^2 - 3a^2) = 0$$

Corollary. The area of the equilateral triangle formed by the pair of lines

$$(ax + by)^2 - 3(bx - ay)^2 = 0 \text{ and the line } ax + by + c = 0 \text{ is } \frac{c^2}{(a^2 + b^2)\sqrt{3}} \text{ sq.units.}$$

Proof. Let p be the altitude from O to $\overline{AB}$ i.e., to the line $ax + by + c = 0$

$$\Rightarrow p = \frac{|c|}{\sqrt{a^2 + b^2}}. \text{ Area of equilateral triangle} = \frac{p^2}{\sqrt{3}} = \frac{c^2}{(a^2 + b^2)\sqrt{3}} \text{ sq.units}$$

Worksheet -3

PAIR OF STRAIGHT LINES

LEVEL-1

SINGLE CORRECT CHOICE TYPE:

- If $kx^2 + 7xy - 3y^2 + 4x + 7y + 2 = 0$ represents a pair of straight lines, then $k =$
 1) 2 2) 3 3) -3 4) -4
- The point of intersection of the pair of lines represented by $xy + x + y + 1 = 0$ is
 1) (1, 1) 2) (-1, 1) 3) (1, -1) 4) (-1, -1)
- The distance between the parallel lines $9x^2 - 6xy + y^2 + 18x - 6y + 8 = 0$ is
 1) $\frac{1}{\sqrt{10}}$ 2) $\frac{2}{\sqrt{10}}$ 3) $\frac{4}{\sqrt{10}}$ 4) $\sqrt{10}$
- If (1, k) is the point of intersection of the lines given by $2x^2 + 5xy + 3y^2 + 6x + 7y + 4 = 0$, then $k =$
 1) 2 2) -2 3) 1 4) -1

MULTI CORRECT CHOICE TYPE:

- The distance between the parallel lines $4x^2 + 2kxy + y^2 + 6x + ky - 10 = 0$ is
 1) $\frac{7}{\sqrt{5}}$ 2) $\sqrt{\frac{49}{5}}$ 3) $\frac{5}{\sqrt{7}}$ 4) $\sqrt{\frac{25}{7}}$

REASONING TYPE:

6. **Statement I:** If $x^2 - y^2 + ky = a^2$ represents a pair of straight lines, then $k = \pm 2a$.
Statement II: If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines, then $abc + 2fgh - af^2 - bg^2 - ch^2 = 0$.
- Both Statements are true, Statement II is the correct explanation of Statement I.
 - Both Statements are true, Statement II is not correct explanation of Statement I.
 - Statement I is true, Statement II is false.
 - Statement I is false, Statement II is true.

COMPREHENSION TYPE:

If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of lines and $h^2 > ab$, then

the point of intersection of the lines is $\left(\frac{hf - bg}{ab - h^2}, \frac{gh - af}{ab - h^2} \right)$.

- The point of intersection of the pair of lines represented by $xy + 2x + 2y + 4 = 0$ is
 1) (2, 2) 2) (-2, 2) 3) (2, -2) 4) (-2, -2)
- The point of intersection of the pair of lines represented by $2x^2 + 5xy + 2y^2 - 17x - 19y + 35 = 0$ is
 1) (3, 1) 2) (-1, 1) 3) (2, 1) 4) (1, 2)
- The point of intersection of the pair of lines represented by $2x^2 - 13xy - 7y^2 + 22x + 41y - 52 = 0$ is
 1) (3, 1) 2) (-1, 1) 3) (2, 1) 4) (1, 2)

MATRIX MATCH TYPE:

10. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ represents a pair of straight lines .
- | | |
|------------------|-------------------------|
| COLUMN -I | COLUMN -II |
| a) If $a = 0$ is | 1) $2fgh = af^2 + ch^2$ |
| b) If $b = 0$ is | 2) $abc = bg^2 + ch^2$ |
| c) If $c = 0$ is | 3) $2fgh = bg^2 + ch^2$ |
| d) If $f = 0$ is | 4) $2fgh = af^2 + bg^2$ |

LEVEL-2**SINGLE CORRECT CHOICE TYPE:**

11. The value if 'k' if the equation $4x^2 + 2hxy + y^2 + 6x + ky - 10 = 0$ represents a pair of parallel lines is
 1) 3 2) 4 3) 2 4) -1

MULTI CORRECT CHOICE TYPE:

12. The value if 'k' if the equation $kx^2 - 10xy + 12xy + 5x - 16y - 3 = 0$ represent a pair of parallel lines is

1) $\frac{5}{2\sqrt{3}}$

2) $\frac{-5}{2\sqrt{3}}$

3) $\frac{5}{\sqrt{3}}$

4) $\frac{2}{3}$

LEVEL-3**SINGLE CORRECT CHOICE TYPE:**

13. The length of the intersect cut off by the pair of lines $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ on the y-axis is

1) $\frac{2\sqrt{g^2 - ac}}{|a|}$

2) $\frac{\sqrt{g^2 - ac}}{|a|}$

3) $\frac{2\sqrt{f^2 - bc}}{|b|}$

4) $\frac{\sqrt{f^2 - bc}}{|b|}$

REASONING TYPE:

14. *Statement I* : The length of the intercept on the x-axis cut by the pair of lines $2x^2 + 5xy + 3y^2 + 6x + 7y + 1 = 0$ is $\sqrt{7}$.

Statement II: The length of the intercept on the x-axis cut by the pair of lines

$$ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0 \text{ is } \frac{2\sqrt{g^2 - ac}}{|a|}.$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

LEVEL-4**SINGLE CORRECT CHOICE TYPE:**

15. If the equation $2xy + gx + fy + 5 = 0$ represent a pair of lines, then $fg =$

1) 1

2) 10

3) 5

4) 0

16. Point of intersection of pair of lines $a(x - \alpha)^2 + 2b(x - \alpha)(y - \beta) + b(y - \beta)^2 = c$ is

1) (0, 0)

2) (α, β)

3) $(-\alpha, -\beta)$

4) $(-\alpha, \beta)$

MATRIX MATCH TYPE:

17. The equation $ax^2 + 2hxy + by^2 + 2gx + 2fx + c = 0$ represents a pair of straight lines .

COLUMN -I

COLUMN -II

a) Intersect on x-axis

1) $h^2 = ab, af^2 = bg^2$

b) Intersect on y-axis

2) $g = f = c = 0, h^2 \geq ab$

c) Intersect at origin

3) $2fgh = bg^2 + ch^2$

d) Never intersect

4) $2fgh = af^2 + ch^2$

LEVEL-5

SINGLE CORRECT CHOICE TYPE:

18. The distance from the point of intersection of the lines $x^2 - y^2 - 2x + 2y = 0$ to the point of intersection of the lines $2x^2 - 5xy + 2y^2 + x + y - 1 = 0$ is
- 1) 1 2) 0 3) 2 4) 4

MATRIX MATCH TYPE:

19. Two pairs of lines

COLUMN -I

COLUMN -II

- | | |
|--|------------------|
| a) $xy = 0$, $xy - 3x - 2y + 6 = 0$ is | 1) Parallelogram |
| b) $6x^2 - 5xy - 6y^2 + x + 5y - 1 = 0$ and $6x^2 - 5xy - 6y^2 = 0$ is | 2) Rhombus |
| c) $2x^2 + 5xy + 2y^2 = 0$, $2x^2 + 5xy + 2y^2 - 15x - 15y + 25 = 0$ is | 3) Square |
| d) $y^2 - 4y + 3 = 0$ and $x^2 + 4xy + 4y^2 + 5x + 10y + 4 = 0$ | 4) Rectangle |

REASONING TYPE:

20. *Statement I* : If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ($c \neq 0$) and $ax^2 + 2hxy + by^2 = 0$ from a parallelogram, then one of the diagonal is $2gx + 2fy + c = 0$.

Statement II : If $ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ ($c \neq 0$) and $ax^2 + 2hxy + by^2 = 0$ from a Rhombus, then $h(f^2 - g^2) + fg(a - b) = 0$.

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

Synopsis -4**PAIR OF STRAIGHT LINES**

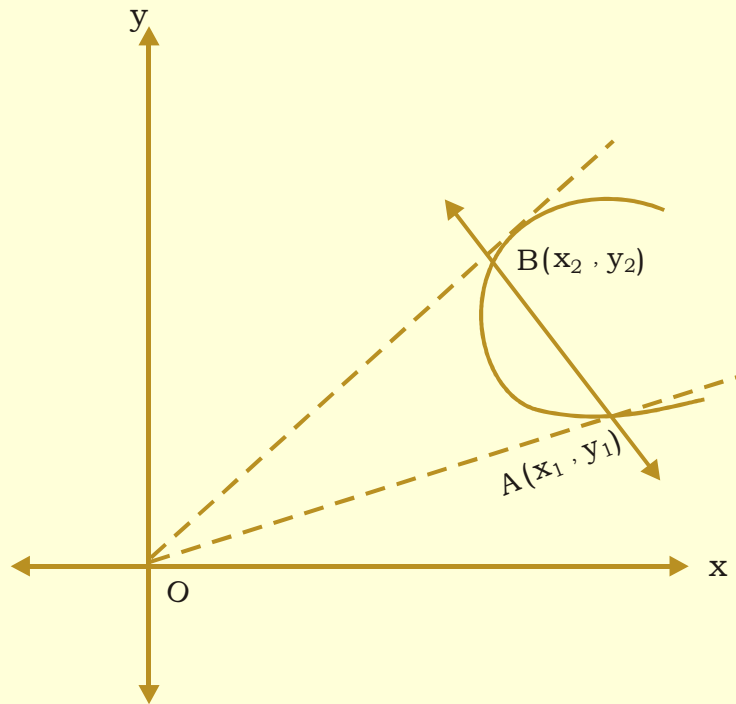
Corollary: The equation of the pair lines joining the origin to the points of intersection of the curve $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and the line $L = lx + my + n = 0$ is

$$ax^2 + 2hxy + by^2 + 2(gx + fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0$$

Proof. Given line is $L = lx + my + n = 0$ i.e., $\frac{lx + my}{-n} = 1$

Given equation of the curve $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$

Let the line $L = 0$ and the curve $S = 0$ meet at the points $A(x_1, y_1)$ and $B(x_2, y_2)$



$$\therefore \frac{lx_1 + my_1}{-x} = 1 \quad \dots\dots\dots(1) \quad \text{and} \quad \therefore \frac{lx_2 + my_2}{-x} = 1 \quad \dots\dots\dots(2)$$

$$\text{Also } ax_1^2 + 2hx_1y_1 + by_1^2 + 2gx_1 + 2fy_1 + c = 0 \quad \dots\dots\dots(3)$$

$$\text{and } ax_2^2 + 2hx_2y_2 + by_2^2 + 2gx_2 + 2fy_2 + c = 0 \quad \dots\dots\dots(4)$$

$$\text{Consider the equation } ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0$$

This is a homogeneous equation and hence represents a pair of straight lines through the origin.

Corollary: The equation to pair of lines passing through (α, β) and parallel to the pair of $ax^2 + 2hxy + by^2 = 0$ is $a(x - \alpha)^2 + 2h(x - \alpha)(y - \beta) + b(y - \beta)^2 = 0$.

Proof. Let the equation $ax^2 + 2hxy + by^2 = 0$ represent the lines

$$l_1x + m_1y = 0 \quad \dots\dots(1) \quad l_2x + m_2y = 0 \quad \dots\dots(2)$$

$$\therefore (l_1x + m_1y)(l_2x + m_2y) = ax^2 + 2hxy + by^2$$

Equating the coefficients of like terms $l_1l_2 = a$, $m_1m_2 = b$, $l_1m_2 + l_2m_1 = 2h$

Now the equation to the lines passing through (α, β) and parallel to (1) and (2) are

$$l_1(x - \alpha) + m_1(y - \beta) = 0 \quad \dots\dots(3) \quad \text{and} \quad l_2(x - \alpha) + m_2(y - \beta) = 0$$

Hence the equation to pair of lines passing through (α, β) and parallel to (1) and

$$(2) \text{ is } [l_1(x - \alpha) + m_1(y - \beta)][l_2(x - \alpha) + m_2(y - \beta)] = 0$$

$$\Rightarrow l_1 l_2 (x - \alpha)^2 + (l_1 m_2 + l_2 m_1)(x - \alpha) + (y - \beta) + m_1 m_2 (y - \beta)^2 = 0$$

$$\Rightarrow a(x - \alpha)^2 + 2h(x - \alpha)(y - \beta) + b(y - \beta)^2 = 0$$

Ex. Find the equation to the pair of lines passing through (1, 2) and parallel to $2x^2 - xy - y^2 = 0$.

Sol. Required equation is $2(x - 1)^2 - (x - 1)(y - 2) - (y - 2)^2 = 0$,
 $2x^2 - xy - y^2 - 2x - 3y - 4 = 0$

Corollary: The equation to the pair of lines passing through the origin and perpendicular to the pair $x^2 + 2hxy + by^2 = 0$ is $bx^2 - 2hxy + ay^2 = 0$.

Proof. Let the lines be $l_1x + m_1y = 0 \quad \dots\dots(1) \quad l_2x + m_2y = 0 \quad \dots\dots(2)$

$$\text{Hence } (l_1x + m_1y)(l_2x + m_2y) \equiv ax^2 + 2hxy + by^2;$$

$$l_1 l_2 = a, \quad m_1 m_2 = b \quad \text{and} \quad l_2 m_2 + l_1 m_1 = 2h$$

Equation to the lines passing through (0, 0) and $\perp$ r to (1) and (2) are

$$mx - l_1y = 0, \quad m_2x - l_2y = 0$$

$$\therefore \text{Equation to pair of perpendicular lines is } (m_1x - l_1y)(m_2x - l_2y) = 0$$

$$\Rightarrow m_1 m_2 x^2 - xy(l_1 m_2 + l_2 m_1) + l_1 l_2 y^2 = 0 \Rightarrow bx^2 - 2hxy + ay^2 = 0$$

Corollry. The equation of the pair of lines through (α, β) and perpendicular to the pair of lines $ax^2 + 2hxy + by^2 = 0$ is $b(x - \alpha)^2 (y - \beta) + a(y - \beta)^2 = 0$

Proof. Let the equation $ax^2 + 2hxy + by^2 = 0$

$$\therefore l_1 l_2 = a, \quad m_1 m_2 = b, \quad l_1 m_2 + l_2 m_1 = 2h$$

Equations to the lines passing through (0, 0) and $\perp$ r to (1) and (2) are

$$m_1x - l_1y = 0 \quad \dots\dots(3) \quad m_2x - l_2y = 0 \quad \dots\dots(3)$$

$\therefore$ Equations to the lines passing through (α, β) and parallel to (3) and (4) is

$$m_1(x - \alpha) - l_1(y - \beta) = 0 \quad \dots\dots\dots(5) \quad m_2(x - \alpha) - l_2(y - \beta) = 0 \quad \dots\dots\dots(6)$$

Hence the equation to the pair lines passing through (α, β) and parallel to (3) and

$$(4) \text{ is } [m_1(x - \alpha) - l_1(y - \beta)][m_2(x - \alpha) - l_2(y - \beta)] = 0$$

$$\Rightarrow m_1 m_2 (x - \alpha)^2 - (l_1 m_2 + l_2 m_1)(x - \alpha)(y - \beta) + l_1 l_2 (y - \beta)^2 = 0$$

$$\Rightarrow b(x - \alpha)^2 - 2h(x - \alpha)(y - \beta) + a(y - \beta)^2 = 0$$

Hence the equation to the pair of lines passing through (α, β) and perpendicular to (1) and (2) is $b(x - \alpha)^2 - 2h(x - \alpha)(y - \beta) + a(y - \beta)^2 = 0$.

Worksheet -4

PAIR OF STRAIGHT LINES

LEVEL-1

SINGLE CORRECT CHOICE TYPE:

1. To find the angle between the pair of lines joining the origin to the points of intersection of the line and the curve, we use the formula.

$$1) \cos \theta = \frac{a + b}{\sqrt{(a + b)^2 + 2h^2}}$$

$$2) \cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + 2h^2}}$$

$$3) \cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + 4h^2}}$$

$$4) \cos \theta = \frac{a + b}{\sqrt{(a - b)^2 + h^2}}$$

2. If the angle between the pair of lines joining the origin to the points of intersection of the curve and the line in $ax^2 + 2hx + by^2 = 0$ is 90° , then
 1) $a + b > 0$ 2) $a + b < 0$ 3) $a + b = 0$ 4) None of these
3. If the angle between the pair of lines joining the origin to the points of intersection of the curve and the line is $\frac{\pi^c}{2}$, then $\cos \theta =$
 1) -1 2) 1 3) 0 4) 2

Multi Correct Choice Type:

4. If the lines joining the origin to the points of intersection of the line $x + 2y = k$ with the curve $2x^2 - 2xy + 3y^2 + 2x - y - 1 = 0$ are at right angles then the value of k is
 1) $k = 1$ 2) $k = 2$ 3) $k = -2$ 4) $k = -1$

REASONING TYPE:

5. *Statement I* : The equation of the pair of lines joining the origin to the point of intersection of the curve $x^2 + y^2 + 3y + 1 = 0$ and the line $x + y - 1 = 0$ is $3x^2 + 5y^2 + 8xy = 0$.
- Statement II* : The equation of the pair of lines joining the origin to the points of

intersection of the curve $S = ax^2 + 2hxy + by^2 + 2gx + 2fy + c = 0$ and the line

$$lx + my + n = 0. \quad ax^2 + 2hxy + by^2 + (2gx + 2fy)\left(\frac{lx + my}{-n}\right) + c\left(\frac{lx + my}{-n}\right)^2 = 0.$$

- Both Statements are true, Statement II is the correct explanation of Statement I.
- Both Statements are true, Statement II is not correct explanation of Statement I.
- Statement I is true, Statement II is false.
- Statement I is false, Statement II is true.

COMPREHENSION TYPE:

$kx + hy = 2hk$ represents a straight line $(x - h)^2 + (y - k)^2 = c^2$ represents a curve.

- The simplified form of $(x - h)^2 + (y - k)^2 = c^2$ is
 - $x^2 + h^2 + y^2 + k^2 = 2xh + 2yh + c^2$
 - $2xh + 2yh + c^2 = 0$
 - $x^2 + h^2 - y^2 - k^2 = 2xh + 2yk + c^2$
 - $x^2 - h^2 + y^2 + k^2 = 2xh - 2yk - c^2$
- The straight line joining the origin to the point of intersection of the straightline $kx + hy = 2hk$ with the curve $x^2 + y^2 - 2xh - 2yk + h^2 + k^2 - c^2 = 0$

$$\begin{aligned}
 &1) \quad x^2 + y^2 - x \frac{(kx + hy)}{k} - y \frac{(kx + hy)}{h} & 2) \quad + (h^2 + k^2 - c^2) \frac{(kx + hy)^2}{2hk} \\
 &3) \quad x^2 + y^2 + x \frac{(kx + hy)}{k} + y \frac{(kx + hy)}{h} & 4) \quad + (h^2 + k^2 + c^2) \left(\frac{kx + hy}{2hk} \right)^2 = 0
 \end{aligned}$$

- If the straight line joining the origin to the points of intersection of the straight line with the curve are at right angles
 - $\cos \theta = 0$
 - coefficient of x^2 + coefficient of $y^2 = 0$
 - $\cos \theta = \frac{1}{2}$
 - (1) and (2)

MATRIX MATCH TYPE:

9. COLUMN -I

COLUMN -II

1) $\cos \theta = \frac{1}{2}$ then $b =$

p) $x = 30^\circ$

2) $\cos x = \frac{\sqrt{3}}{2}$ then $x =$

q) $y = 45^\circ$

3) $\cos y = \frac{1}{\sqrt{2}}$ then $a =$

r) $a = 90^\circ$

4) $\cos a = 0$ then $a =$

s) $b = 60^\circ$

LEVEL-2**SINGLE CORRECT CHOICE TYPE:**

10. Equation of pair of lines passing through the origin and point of intersection of $x^2 + 2y^2 = 4$ and $x + y = 1$ is
- 1) $3x^2 - 8xy + 3y^2 = 0$ 2) $3x^2 + 8xy + 2y^2 = 0$
 3) $2x^2 - 8xy + 3y^2 = 0$ 4) $2x^2 + 8xy + 3y^2 = 0$
11. The lines joining the origin to the points of intersection of $2x^2 + 2xy - 4x + 1 = 0$ and $3x + y = 1$ are given by
- 1) $x^2 - y^2 - 4xy = 0$ 2) $x^2 - y^2 + 5xy = 0$
 3) $x^2 + y^2 - 5xy = 0$ 4) $x^2 + y^2 + 5xy = 0$

REASONING TYPE:

12. *Statement I* : The straight line $y + tx = 2at + at^3$ cuts the curve $y^2 - 4ax = 0$ at A and B. If AB subtends a right angle at the origin the gradient of AB is $\pm\sqrt{2}$.
- Statement II* : If two lines represented by equation has coefficient of $x^2 +$ coefficient of $y^2 = 0$. Then the lines are mutually perpendicular.
1. Both Statements are true, Statement II is the correct explanation of Statement I.
 2. Both Statements are true, Statement II is not correct explanation of Statement I.
 3. Statement I is true, Statement II is false.
 4. Statement I is false, Statement II is true.

LEVEL-3**SINGLE CORRECT CHOICE TYPE:**

13. The equation to the pair of straight lines joining the origin to the intersection points of the straight line $y + mx = c$ and the curve $x^2 + y^2 = a^2$ is represented by $S=0$. If the angle between the pairs of lines represented $S = 0$ is 90° then
- 1) $c^2 = a^2(1 + m^2)$ 2) $c^2 = a^2(1 - m^2)$ 3) $2c^2 = a^2(1 + m^2)$ 4) $2c^2 = a^2(1 - m^2)$

Multi Correct Choice Type:

14. If the straight line joining the origin to the intersection points of the straight line $ax + by - 2ab = 0$ and the curve $(x - b)^2 + (y - a)^2 = c^2$ are at right angle. then
- 1) $a^2 = (c + b)(c - b)$ 2) $b^2 = (c + a)(c - a)$
 3) $a^2 + b^2 = c^2$ 4) $a^2 + b^2 + c^2 = 0$

LEVEL-4**SINGLE CORRECT CHOICE TYPE:**

15. The pair of straight lines joining the origin to the common points of $x^2 + y^2 = 4$ and $y = 3x + c$ are perpendicular if $c^2 =$
- 1) -1 2) 6 3) 13 4) 20

MATRIX MATCH TYPE:

16. If the equation to the pair of lines joining the origin to the points of intersection of $2x + 3y = 1$ and $x^2 + y^2 = 4$ is $ax^2 + bxy + cy^2 = 0$, then

COLUMN -I

COLUMN -II

1) a

p) 35

2) b

q) 48

3) c

r) 15

4) $a + b + c$

s) 99

t) 98

LEVEL-5**COMPREHENSION TYPE:**

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ represents a curve $lx + my + n = 0$ represents a straight line

17. The simplified given of $lx + my + n = 0$ is

1) $\frac{lx - my}{-n} = 1$

2) $\frac{lx + my}{-n} = 0$

3) $\frac{lx + my}{-n} = 1$

4) $\frac{lx + my}{n} = 1$

18. The equation of the lines joining the origin to the point of intersection of $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ with $lx + my + n = 0$ is

1) $\frac{x^2(n^2 - l^2)}{a^2} + y^2 \frac{(n^2 - m^2)}{b^2} + 2lmxy = 0$

2) $\frac{x^2(n^2 + l^2)}{a^2} + y^2 \frac{(n^2 + m^2)}{b^2} - 2lmxy = 0$

3) $\frac{x^2(n^2 - l^2)}{a^2} + y^2 \frac{(n^2 - m^2)}{b^2} - 2lmxy = 0$

4) None of these

19. If the pair of lines joining the origin to the intersection of the curve $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ by the line $lx + my + n = 0$ are coincident if

1) $a^2l^2 - b^2m^2 = n^2$

2) $a^2l^2 + b^2m^2 + n^2 = 0$

3) $a^2l^2 + b^2m^2 = b^2$

4) None